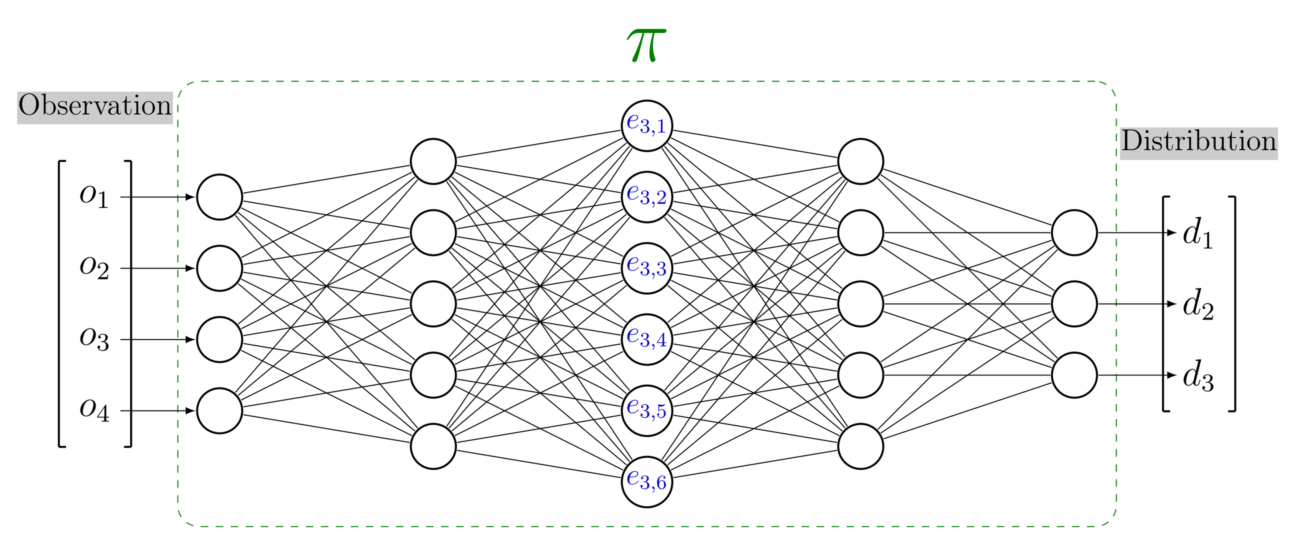
Clustering agent representation for explaining reinforcement learning policies

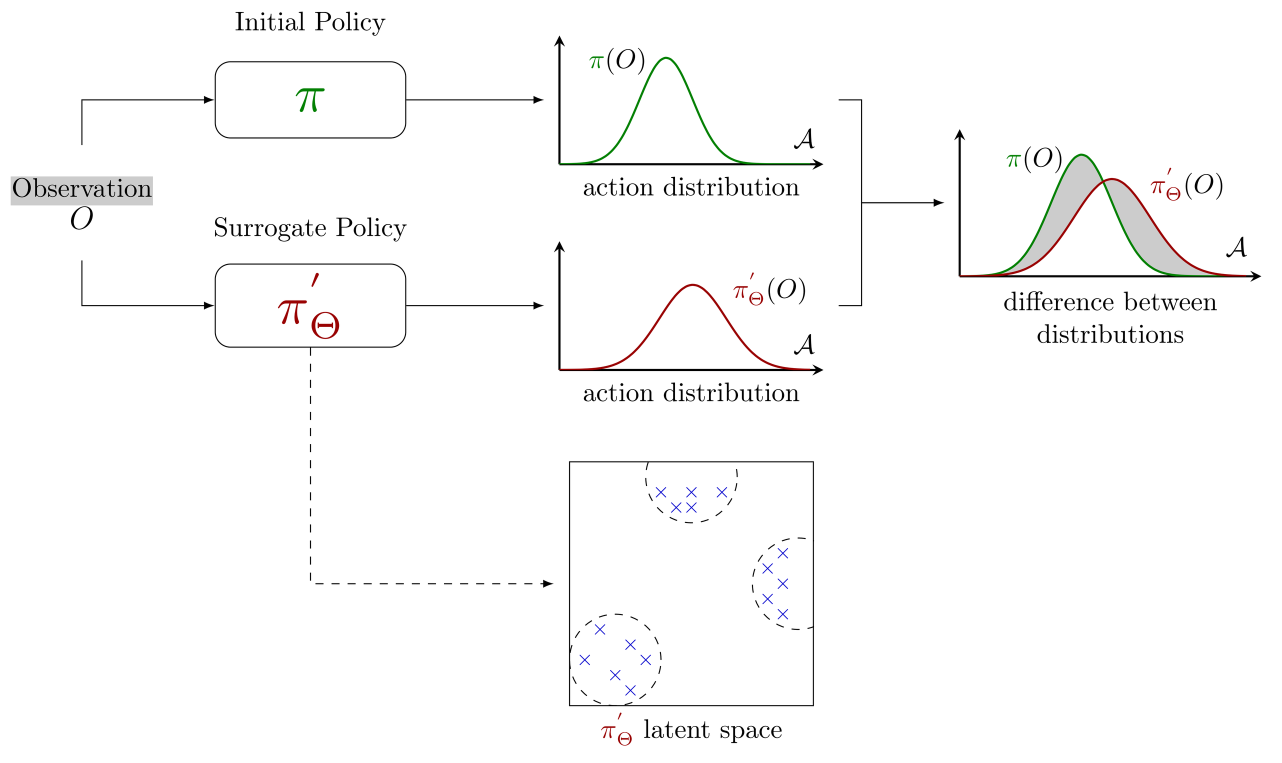
Maxime Alaarabiou • Nicolas Delestre • Laurent Vercouter

INSA Rouen Normandie, Normandie Univ, LITIS UR 4108, F-76000, 684 Av. de l'Université, 76800 Saint-Étienne-du-Rouvray, France

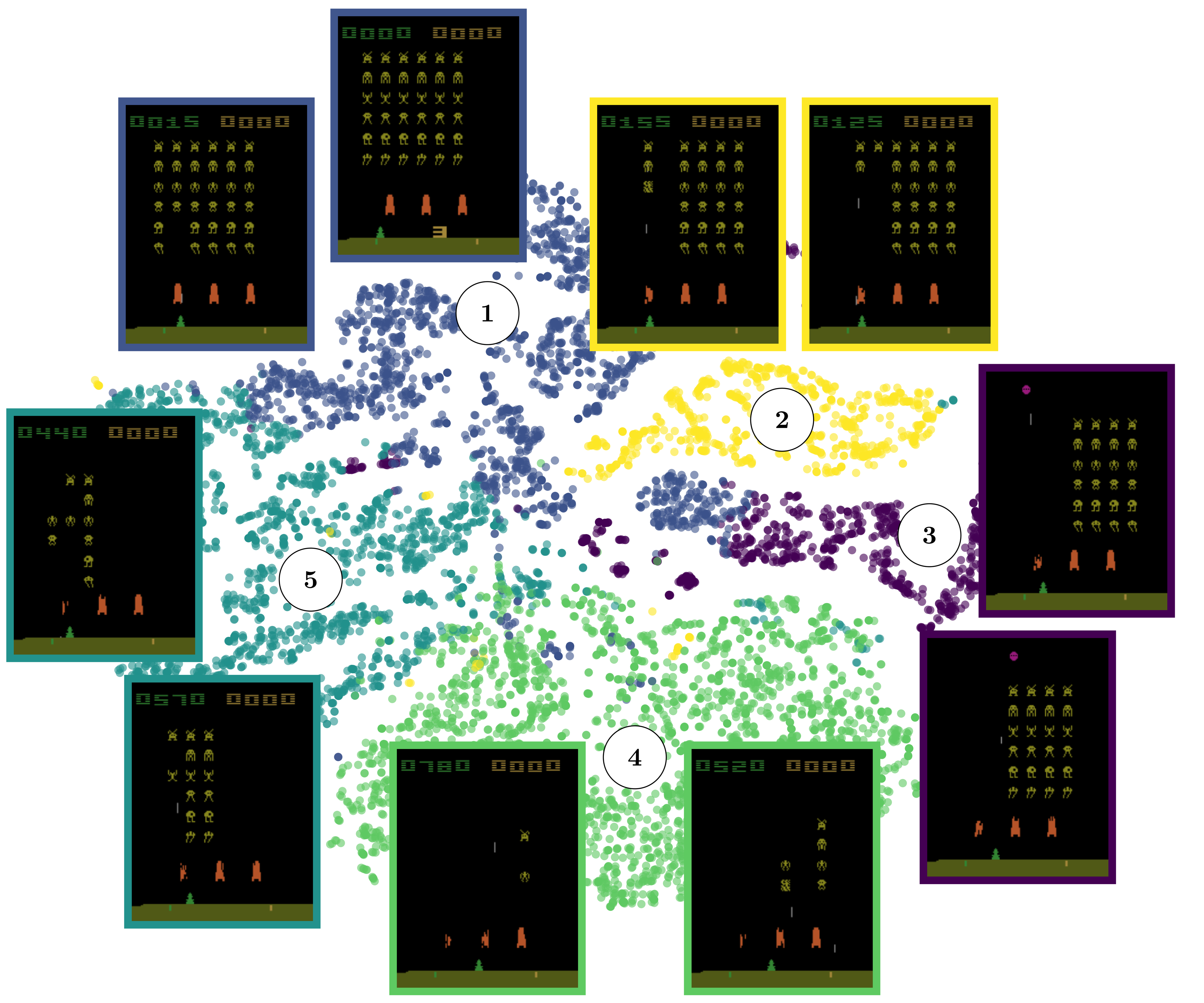
Latent Space



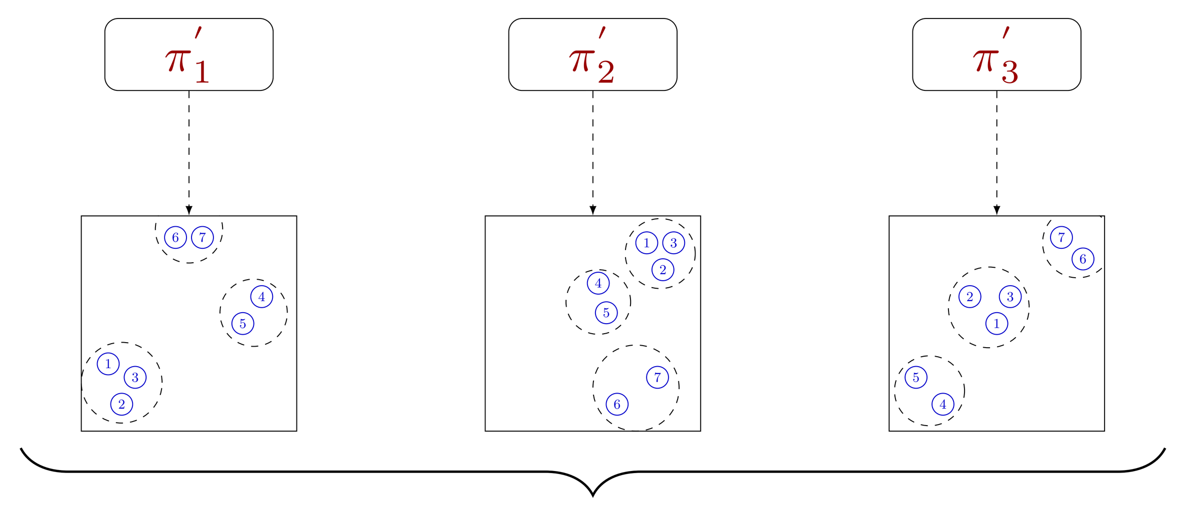
Clustering Agent Representation



Semantic Latent Space

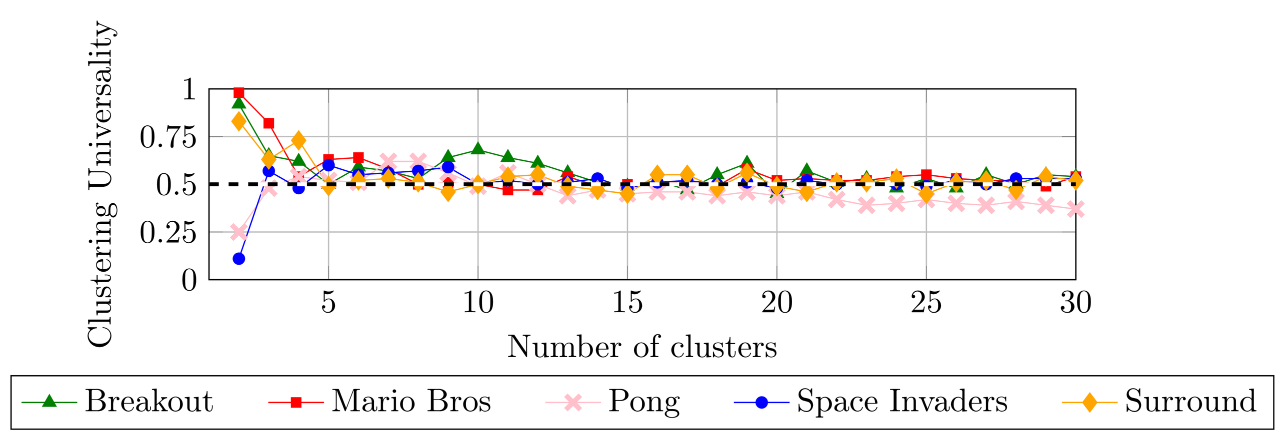


Clustering Universality



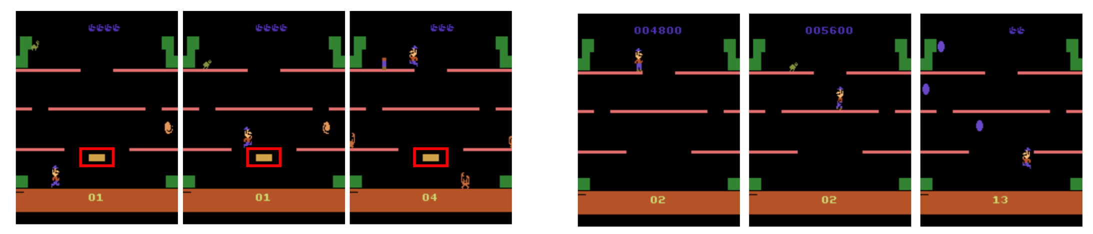
$$\textit{Clustering Universality} = \frac{1}{\binom{n}{2}} \sum_{i=1}^{n} \sum_{j=0}^{i-1} \mathrm{ARI}(\mathcal{C}_i, \mathcal{C}_j)$$

Evolution Clustering Universality



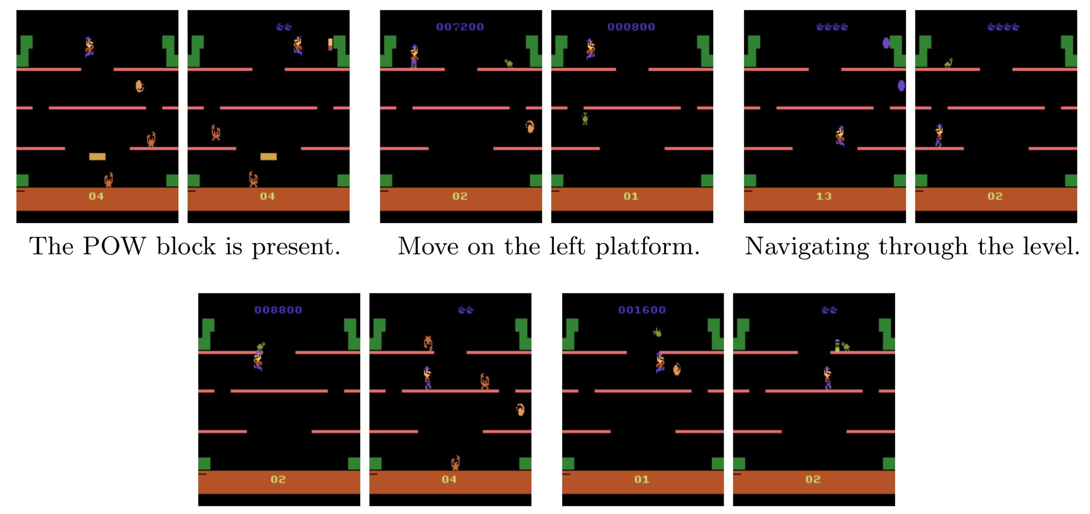
Semantic Granularity

The case with 2 clusters.



The POW block is present. The POW block is absent.

The case with 5 clusters.



The POW block is present. Move on the left platform. Navigating through the level.

Attacking the left. Attacking the right.